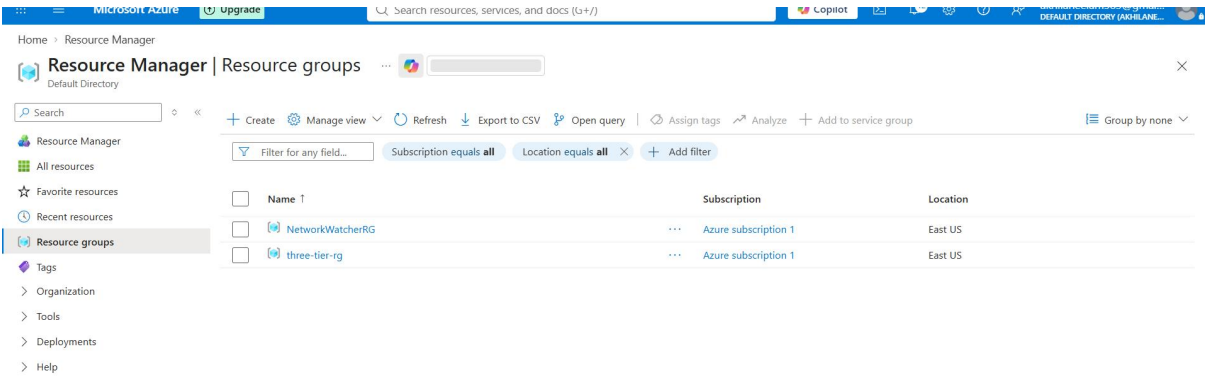


3 TIER ARCHITECTURE

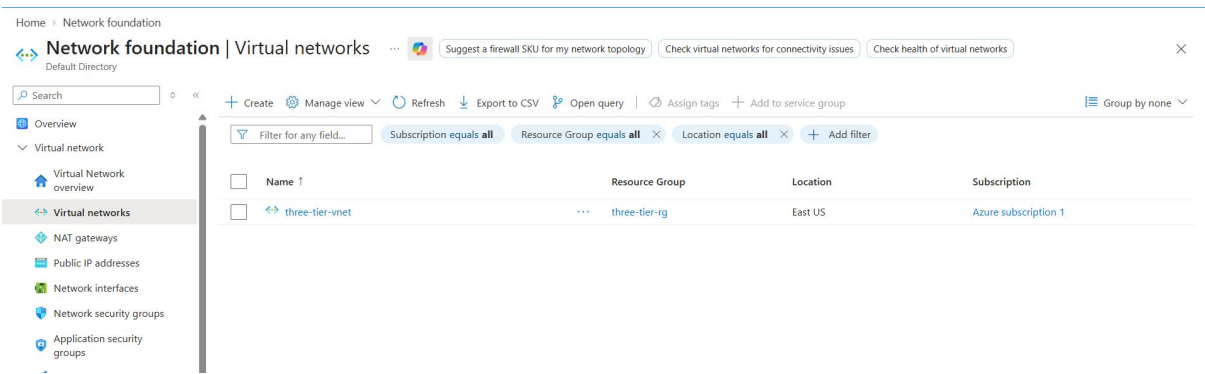
Name:Neelam Akhila

Batch No:44

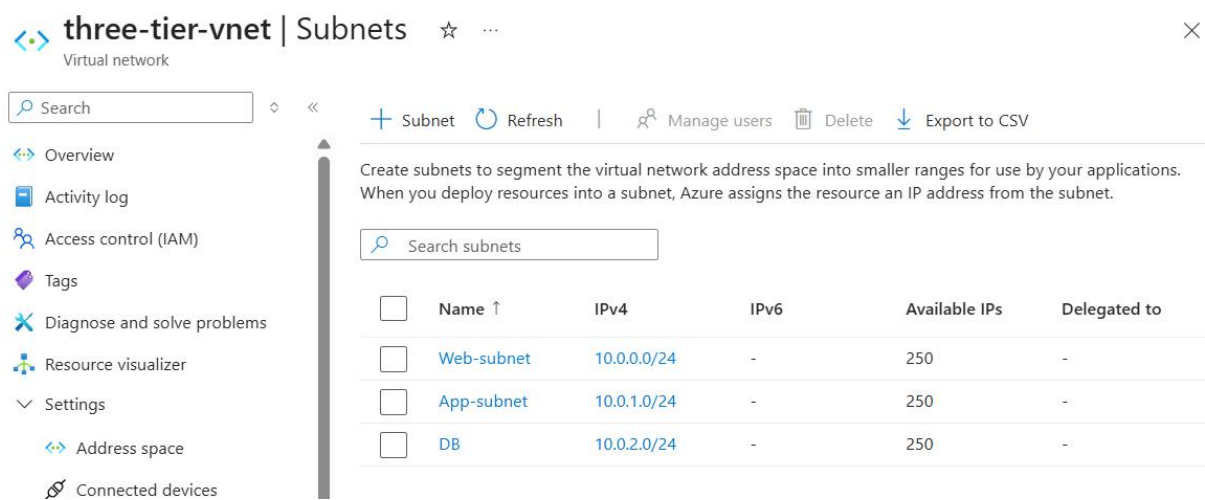
1.Azure Resource Group Creation



2.Virtual Network



3.Created three subnets

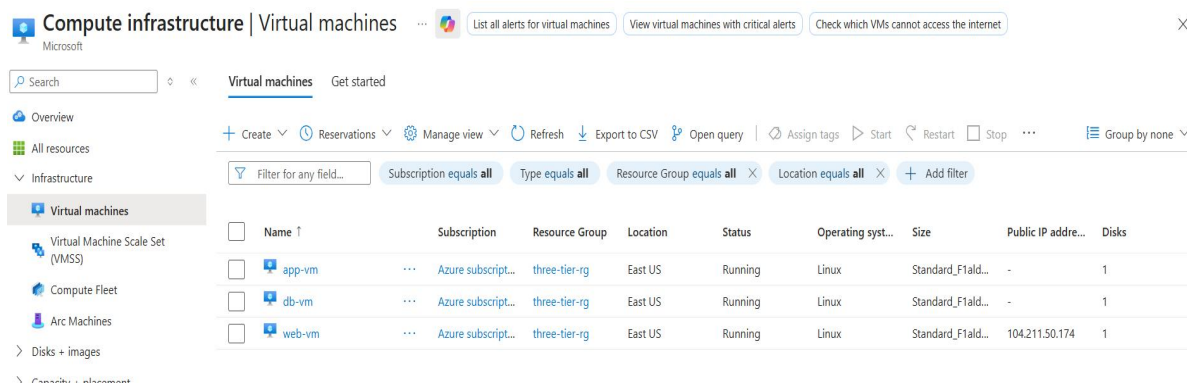


4. Created Three virtual Machines:Web,APP,DB

I created three virtual machines (VMs) for the three-tier architecture: Web VM, Application VM, and Database VM.

The Web VM was assigned a Public IP address for Internet access, while the Application VM and Database VM were kept private without Public IP addresses for better security.

The Web VM was configured with **SSH (port 22)** for secure remote administration and **HTTP (port 80)** to allow users to access the web application through its public IP address.



The screenshot shows the Azure portal interface for 'Virtual machines'. It lists three VMs: 'app-vm', 'db-vm', and 'web-vm'. The 'web-vm' has a public IP address of 104.211.50.174, while the others do not. All VMs are running Linux and are part of the 'three-tier-rg' resource group in the 'East US' location.

Name	Subscription	Resource Group	Location	Status	Operating system	Size	Public IP address	Disks
app-vm	Azure subscri...	three-tier-rg	East US	Running	Linux	Standard_F1ald...	-	1
db-vm	Azure subscri...	three-tier-rg	East US	Running	Linux	Standard_F1ald...	-	1
web-vm	Azure subscri...	three-tier-rg	East US	Running	Linux	Standard_F1ald...	104.211.50.174	1

5.Installation and Configuration of Nginx Web Server

```
63 packages can be upgraded. Run 'apt list --upgradable' to see them.
root@web-vm:/home/Akhila# apt install nginx -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  nginx-common
Suggested packages:
  fcgiwrap nginx-doc ssl-cert
The following NEW packages will be installed:
  nginx nginx-common
0 upgraded, 2 newly installed, 0 to remove and 63 not upgraded.
Need to get 579 kB of archives.
After this operation, 1,121 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://deb.debian.org/debian bullseye/main amd64 nginx-common all 1:1.20.1-1~deb11u1 [16.1 kB]
Get:2 http://deb.debian.org/debian bullseye/main amd64 nginx amd64 1:1.20.1-1~deb11u1 [563 kB]
Fetched 579 kB in 0s (10.4 MB/s)
Selecting previously unselected package nginx-common.
(Reading database ... 123456 files and directories currently installed.)
Preparing to unpack .../nginx-common_1:1.20.1-1~deb11u1_all.deb ...
Unpacking nginx-common (1:1.20.1-1~deb11u1) ...
Selecting previously unselected package nginx.
Preparing to unpack .../nginx_1:1.20.1-1~deb11u1_amd64.deb ...
Unpacking nginx (1:1.20.1-1~deb11u1) ...
Setting up nginx-common (1:1.20.1-1~deb11u1) ...
Setting up nginx (1:1.20.1-1~deb11u1) ...
invoke-rc.d: could not determine current runlevel
invoke-rc.d: policy-rc.d denied execution of start.
root@web-vm:/home/Akhila# cd /var/www/html/
root@web-vm:/var/www/html# ls
index.nginx-debian.html
root@web-vm:/var/www/html# vi index.nginx-debian.html
```

6.SSH Connectivity Testing using Telnet

Telnet was used to verify **SSH connectivity** between the Web VM and Application VM using the Application VM's **private IP address (10.0.1.4)** on **port 22**.

The successful connection confirmed that the Web VM can securely communicate with the Application VM through the **private network**.

```
root@web-vm:/var/www/html# telnet 10.0.1.4 22
Trying 10.0.1.4...
Connected to 10.0.1.4.
Escape character is '^]'.
SSH-2.0-OpenSSH_9.6p1 Ubuntu-3ubuntu13.19
```

After successfully installing Nginx, the Web VM was accessed through its Public IP address in a web browser. The default Nginx welcome page was displayed, confirming that the Nginx web server was successfully installed and running.

Welcome to Web server!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

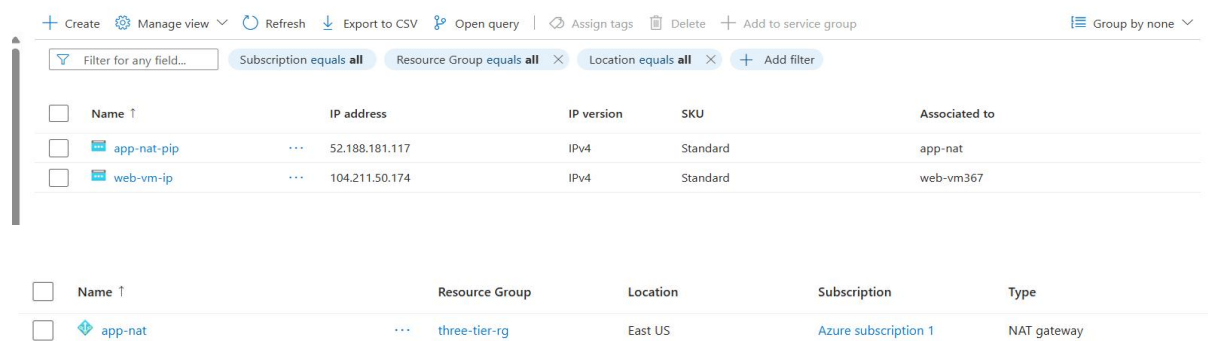
For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

7. NAT Gateway Configuration

Since the Application and Database VMs do not have Public IP addresses, a NAT Gateway was created to provide secure outbound Internet connectivity. A Public IP was associated with the NAT Gateway, and the NAT Gateway was attached to both the Application and Database subnets.

Outbound rules were configured to allow the required Internet access for the Application and Database VMs, while keeping both VMs private and inaccessible directly from the Internet.



	Name ↑	IP address	IP version	SKU	Associated to
☐	app-nat-pip	52.188.181.117	IPv4	Standard	app-nat
☐	web-vm-ip	104.211.50.174	IPv4	Standard	web-vm367

	Name ↑	Resource Group	Location	Subscription	Type
☐	app-nat	three-tier-rg	East US	Azure subscription 1	NAT gateway

First, we tested connectivity from the Web VM to the Application VM using the private IP 10.0.1.4 on port 8080, but the connection was refused because Tomcat was not installed and running yet.

```
root@web-vm:/var/www/html# telnet 10.0.1.4 8080
Trying 10.0.1.4...
telnet: Unable to connect to remote host: Connection refused
root@web-vm:/var/www/html# ssh Akhila@10.0.1.4
The authenticity of host '10.0.1.4 (10.0.1.4)' can't be established.
ED25519 key fingerprint is SHA256:NAbhMTdSpu0WRJnpKtQ/us3ak7aLqYiczA3dgsdZJaE.
This key is not known by any other names.
```

Then, we connected to the Application VM using **SSH (port 22)** with its private IP:

```
root@web-vm:/var/www/html# ssh Akhila@10.0.1.4
The authenticity of host '10.0.1.4 (10.0.1.4)' can't be established.
ED25519 key fingerprint is SHA256:NAbhMTdSpu0WRJnpKtQ/us3ak7aLqYiczA3dgsdZJaE.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.0.1.4' (ED25519) to the list of known hosts
```

For the Application VM, inbound access was configured for SSH (Port 22) and Tomcat (Port 8080) to enable secure access and application communication.

Inbound port rules (5)						
100	Allow-ssh		22	TCP	10.0.0.4	
110	Allow-tomcat		8080	Any	10.0.0.4	
65000	AllowVnetInBound ⓘ		Any	Any	VirtualNetwork	
65001	AllowAzureLoadBalancerInB... ⓘ		Any	Any	AzureLoadBala	
65500	DenyAllInBound ⓘ		Any	Any	Any	
Outbound port rules (3)						

```
sudo apt update
```

```
sudo apt install -y openjdk-17-jdk
```

```
sudo apt install -y tomcat10
```

```
systemctl start tomcat10
```

```
systemctl status tomcat10
```

```
Akhila@app-vm:~$ sudo systemctl status tomcat10
● tomcat10.service - Apache Tomcat 10 Web Application Server
   Loaded: loaded (/usr/lib/systemd/system/tomcat10.service; enabled; preset:
   Active: active (running) since Sun 2026-10-04 13:39:16 UTC; 25s ago
     Docs: https://tomcat.apache.org/tomcat-10.0-doc/index.html
   Process: 4851 ExecStartPre=/usr/libexec/tomcat10/tomcat-update-policy.sh (c
   Main PID: 4856 (java)
    Tasks: 30 (limit: 2245)
   Memory: 69.4M (peak: 71.4M)
      CPU: 1.952s
   CGroup: /system.slice/tomcat10.service
           └─4856 /usr/lib/jvm/java-17-openjdk-amd64/bin/java -Djava.util.log

Oct 04 13:39:17 app-vm tomcat10[4856]: OpenSSL successfully initialized [OpenSS
Oct 04 13:39:17 app-vm tomcat10[4856]: Initializing ProtocolHandler [http-nio-
Oct 04 13:39:17 app-vm tomcat10[4856]: Server initialization in [1119] millisec
Oct 04 13:39:17 app-vm tomcat10[4856]: Starting service [Catalina]
Oct 04 13:39:17 app-vm tomcat10[4856]: Starting Servlet engine: [Apache Tomcat/
```

After successfully installing and configuring Tomcat 10 on the Application VM, the Web VM was used to test the connection to Tomcat through the Application VM's private IP address (10.0.1.4) on port 8080.

The successful connection confirmed that the Web Server can communicate with the Application Server through the private network.

```
Akhila@web-vm:~$ telnet 10.0.1.4 8080
Trying 10.0.1.4...
Connected to 10.0.1.4.
```

After completing the Application Server configuration, I logged in to the **Database VM** using its private network connection and installed the **MySQL Server**

```
Akhila@web-vm:~$ ssh Akhila@10.0.2.4
Akhila@10.0.2.4's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-1022-azure x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Custom information as of Sun Oct 4 14:44:08 UTC 2025
```

Inbound port rules (3)

100	Allow-mysql		3306	TCP	10.0.1.4
110	deny-web		Any	Any	10.0.0.4

```
sudo apt install mysql-server -y
```

```
sudo ss -ltnp | grep 3306
```

```
sudo vi /etc/mysql/mysql.conf.d/mysqld.cnf
```

```
sudo systemctl restart mysql
```

```
sudo ss -ltnp | grep 3306
```

```
bind-address = 0.0.0.0
```

```
mysqlx-bind-address = 127.0.0.1
```

```
systemctl restart mysql
```

Inbound rules were configured on the Database VM to allow MySQL access only from the Application VM. Direct access from the Web VM to the Database VM was restricted to prevent unauthorized or unnecessary communication and to maintain the security of the three-tier architecture.

The connectivity was successfully tested using Telnet on port 3306: the Application VM was able to connect to the Database VM, while the Web VM was denied access.

```
Akhila@app-vm:~$ telnet 10.0.2.4 3306
Trying 10.0.2.4...
Connected to 10.0.2.4.
```

```
Akhila@web-vm:~$ telnet 10.0.2.4 3306
Trying 10.0.2.4...
telnet: Unable to connect to remote host: Connection timed out
```

The three-tier architecture was successfully implemented on Microsoft Azure, with secure communication between the Web, Application, and Database tiers. The project was successfully tested, ensuring that the Web VM can access the Application VM, the Application VM can access the Database VM, and direct Web-to-Database access is restricted.